Jet Engine Low-Fly Detection — Benchmarks

Binary JET vs NO-JET • ESC-50 fold-5 (n=88, 8 jets) + quarantined field clips

Model verdicts

`synth` ● **useless** predicts all-negative; accuracy lies
`esc50` ● **biased** predicts all-jet; no best-checkpoint
`v1` ● **prototype** balanced-ish, crowd slipping
`v2` ● **best so far** top ESC, crowd regressed

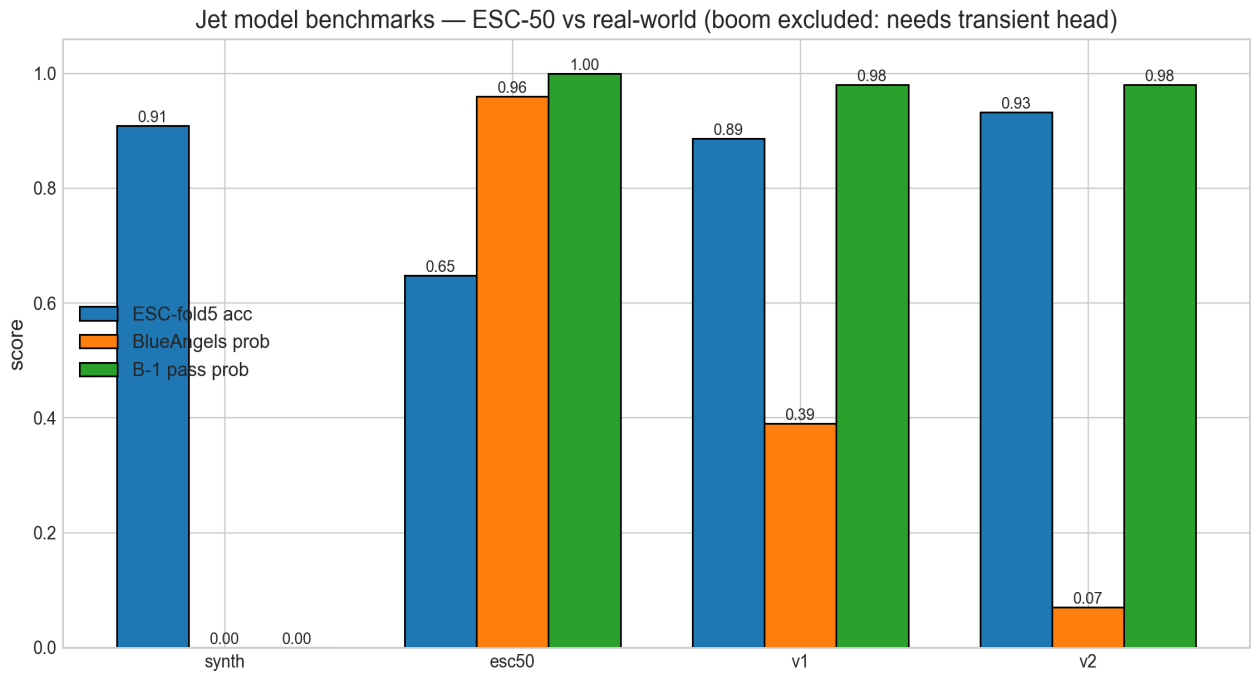
Headline: v2 leads on ESC-50 (0.932 acc, prec 1.0) and nails clean flybys (B-1 0.98), but crowd+music airshow audio is regressing (BlueAngels 0.96 → 0.07).
All four models score 0.00 on sonic booms — a separate 1 s transient detector is required (32 boom clips banked).

Metric table — F1 is the honest column (only 8 jets / 88 clips)

model	ESC acc	prec	rec	F1	BlueAng	B-1	boom
synth	0.909	0.00	0.00	0.000	0.00	0.00	0.00
esc50	0.648	0.21	1.00	0.340	0.96	1.00	0.00
v1	0.886	0.40	0.50	0.444	0.39	0.98	0.00
v2	0.932	1.00	0.25	0.400	0.07	0.98	0.00

green-tinted bests: v2 ESC acc 0.932 / prec 1.00, esc50 BlueAngels 0.96, esc50/v1/v2 B-1 ≥0.98. Raw numbers: benchmarks.csv

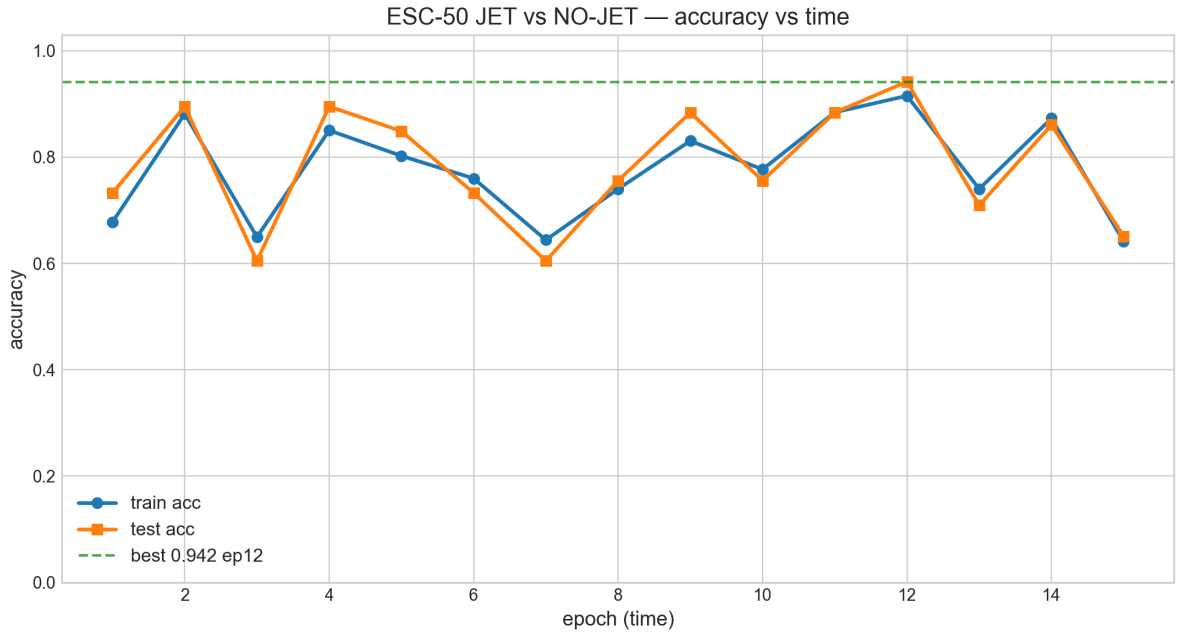
Benchmark chart



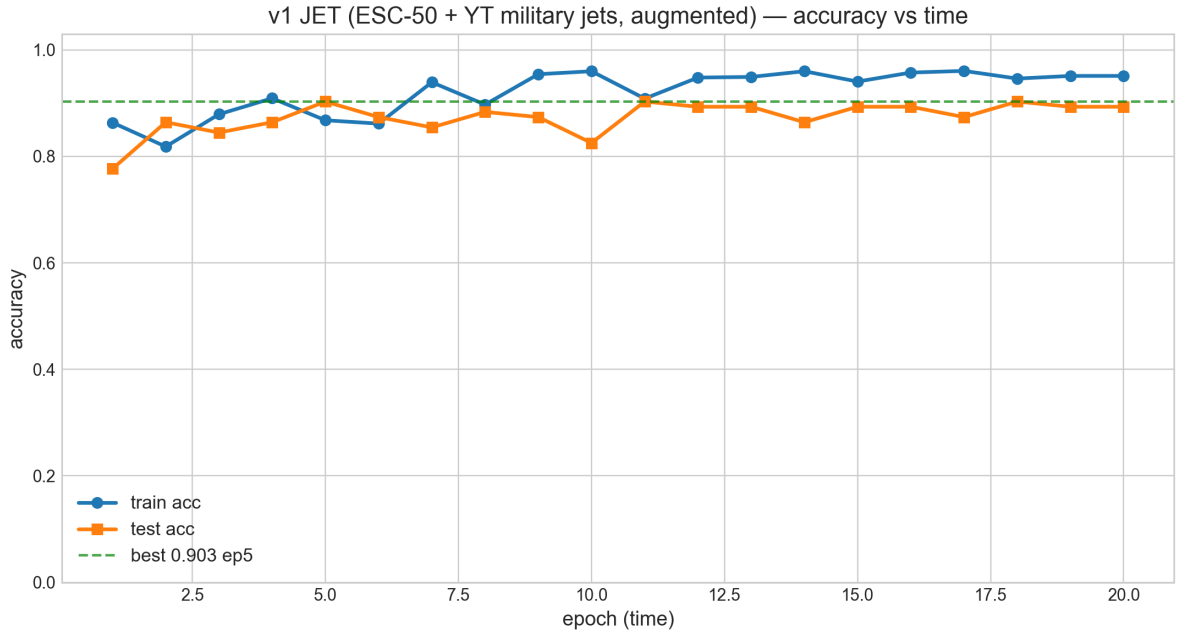
ESC-50 accuracy vs real-world detection probs. Boom excluded (transient head pending).

Learning curves

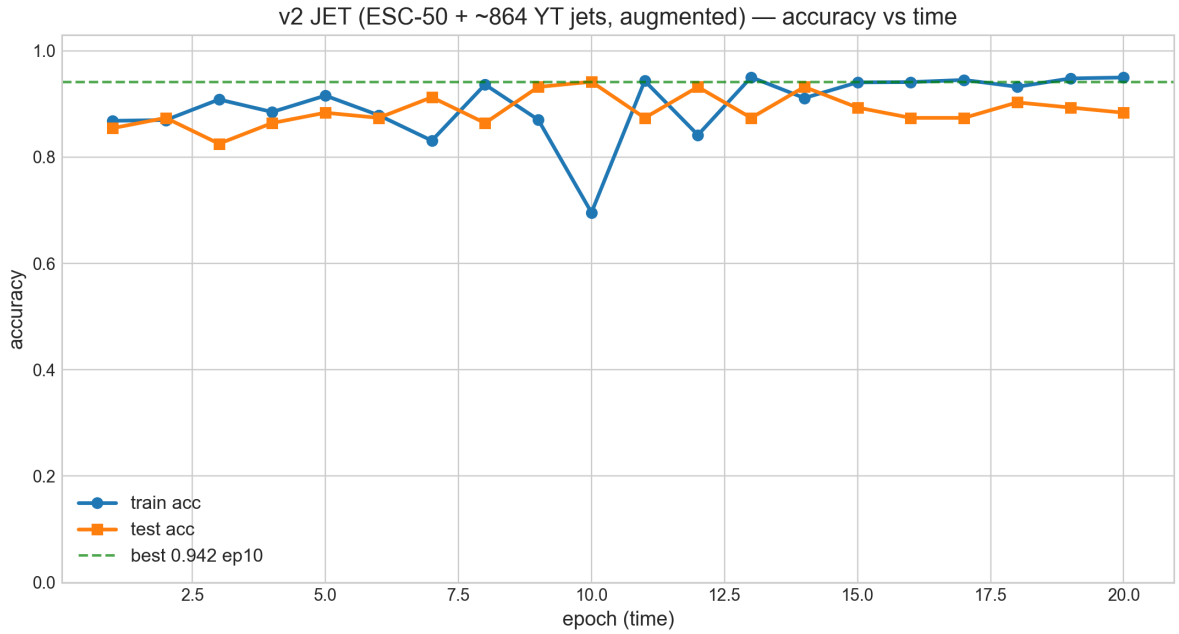
baseline (ESC-50 only): jagged, tiny data



v1 (+YT jets, augment, best-ckpt): stabilised

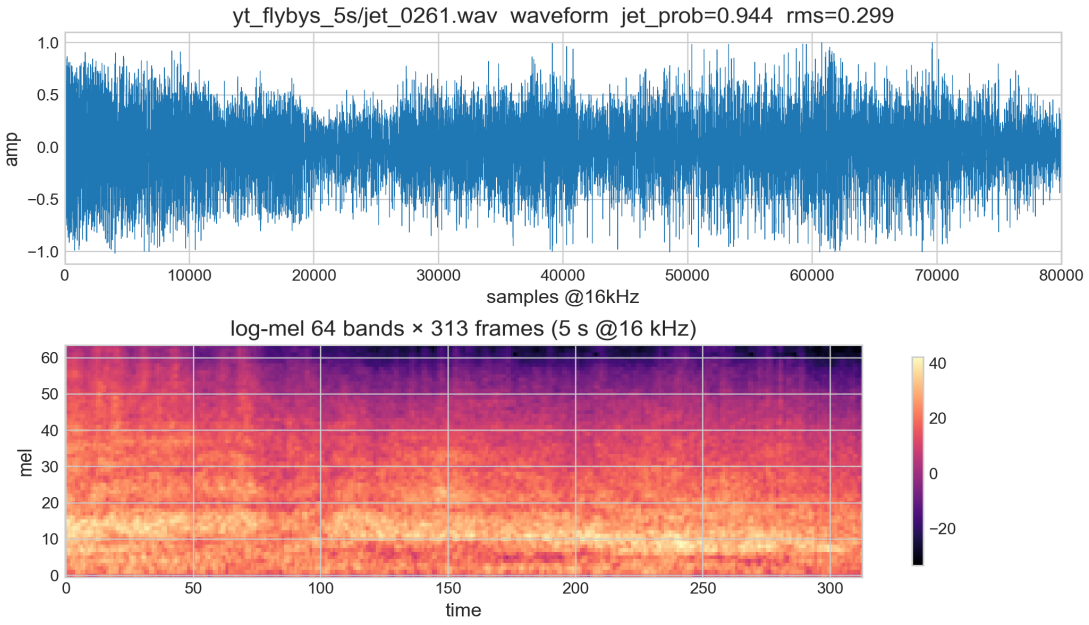


v2 (+~864 YT jets): best 0.942 ep10


v2 JET (ESC-50 + ~864 YT jets, augmented) — accuracy vs time
accuracy
epoch (time)
train acc
test acc
best 0.942 ep10
0.0
0.2
0.4
0.6
0.8
1.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

Sample detection — jet_0261.wav

rms 0.30 (loud) • jet_prob 0.917 • broadband roar, bright low-mel band = classic flyby signature




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Listen: afplay yt_flybys_5s/jet_0261.wav
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Data inventory

split	what	n
JET train	ESC-50 airliners (folds 1–4)	32
JET train	YT compilations 5 s (filtered ≥0.5)	~780
JET train	Abbotsford crowd-jets (thr 0.3, hard kept)	46
JET test-only	B-1 passes, F-16 clips, BlueAngels, Astra	8 files
NO-JET train	ESC-50 hard negs (heli/engine/train/storm…)	~460
NO-JET test	ESC-50 fold-5 negs	~80
BOOM (pending)	2 s transient clips (F-18/F-35/comps)	32
BIG (incoming)	AeroSonicDB v1.1: 470 turbofan + 1270 bg	downloading

Pipeline: 16 kHz mono → 5 s → log-mel(64) → JetCNN-286k → sigmoid. Next: AeroSonicDB retrain on arrival + 1 s boom head.